NOT MEASUREMENT SENSITIVE
MIL-P-85856/15
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS NICKEL-CHROMIUM-BORON, TYPE I, CLASS 1, COMPOSITION N

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Cr	Ni	Fe	C	Si	Co	B
13.0 20.0	Bal	3.0 5.0	0.6 1.3	3.0 5.0	1.0 max.	2.75 4.75

(1) Determination not required for routine acceptance.

Particle Size/Distribution

-200 Mesh (-75 µm)	99.0 min.
+270 Mesh (+53 µm)	20.0 max.

Sprayed Coating Properties

Bond Strength	37.9 MPa (5500 psi) min.
Hardness	Rockwell 15N, 75 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 10% max. Unmelted Particles - 10% max.

AMSC N/A FSC 3439

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Deposit Efficiency and Deposition Rate

85% min for O_2/H_2 @ 20.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

Aerospace Materials Specification AMS 4775

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/15)

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